RUIWEN WANG

AI SYSTEMS | ML SYSTEMS | DISTRIBUTED TRAINING

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Ph.D. candidate (CIFRE) in Computer Science at Sorbonne University and EURECOM, working with Huawei Paris Research Center on efficient and reliable ultra-scale LLM and DNN training. Build symbolic performance and memory models with ILP and constraint-based planners for hybrid parallelism, pipeline scheduling, and recomputation under memory and communication constraints. Doctoral thesis unifies these into one portable planning workflow — predict, jointly optimise, re-plan, and adapt.

Large-scale LLM/DNN training • Symbolic performance and memory modeling • ILP / constraint optimization • Hybrid parallelism and pipeline scheduling

Validated on production Ascend clusters up to **16,384 NPUs** • Best reported speedup **36.72%** • 7 papers (4 first-author) • 3 patent families

EXPERIENCE

Distributed and Parallel Computing Lab, Huawei Paris Research Center
Paris, France

Ph.D. Candidate (CIFRE) and Research Engineer – LLM Training Systems
2021 – end 2026

- Built cost-model-driven planners for hybrid and pipeline-parallel LLM training, covering stage partitioning, pipeline scheduling, and recomputation selection under memory and performance constraints.
- Designed symbolic peak-memory estimators, collective-cost models, and overlap-aware strategies for throughput and MFU tuning at large scale.
- Productionized research prototypes into deployable planning workflows with engineering and customer teams, delivering actionable configuration guidance, debuggable traces, and regression benchmarks across model families and scales.
- Supported technology transfer from research to production-scale training environments, including deployment feedback on customer workloads and long-sequence training scenarios.

Role progression: Research Apprentice (2021–2022) -> Research Engineer (2022–2023) -> CIFRE Ph.D. Candidate (2023–end 2026).

SELECTED SYSTEMS & RESULTS

- **Prism** (HPC Asia 2026) – **predictability via symbolic memory modeling**: profiling-free, layer-level prediction of per-stage peak memory and communication, making memory feasibility a closed-form, retargetable check across model families and clusters.
- **BMPipe** (CLUSTER 2025) – **pipeline efficiency**: symbolic bubble model (recomputation / imbalance / preparation) plus ILP over stage boundaries, chunking, and recomputation; up to 1.36x at 83–92% HBM, validated to 16,384 Ascend NPUs.
- **H2O** (Euro-Par 2025, Best Poster) – **holistic all-dimension optimisation**: co-optimises every parallelism dimension with micro-batch and adaptive recomputation in one workflow; MFU 26.13% to 35.73% on DeepSeek-141B, transferred to production.
- **Concord** (under submission 2026) – **search efficiency**: dependency-keyed per-layer signatures re-plan only what a model or config change invalidates, reusing the rest and cutting redundant search.
- **ManuMatic** (NPC 2025) – **adaptability to new operators**: injects pinned operator shardings as first-class constraints so plans stay robust to unmodeled operators and runtime effects; +30.1% over an expert-tuned plan on Qwen2.5-72B.

KEYWORDS

Hybrid parallelism, pipeline scheduling, MoE / expert parallelism, symbolic cost & peak-memory modeling, ILP / MILP planning, recomputation, communication overlap, MFU optimisation, Ascend NPU, portability / retargeting.

CORE AREAS

- Large-scale LLM and DNN training across tensor, data, pipeline, and expert parallelism
- Symbolic performance and peak-memory modeling for feasibility analysis
- ILP and constraint-based optimization for memory, communication, and scheduling trade-offs
- Pipeline scheduling, stage partitioning, micro-batch planning, and recomputation strategies
- Production-scale validation on GPU/NPU clusters and deployable benchmarking workflows
- Research-to-production transfer with engineering, platform, and customer teams

SELECTED PUBLICATIONS

- **BMPipe** – IEEE CLUSTER 2025
- **H2O** – Euro-Par 2025, Best Poster Award
- **ManuMatic** – NPC 2025
- **PRISM & MLLM Pipeline Bubble Modeling** – HPC Asia 2026
- **Concord:** dependency-keyed re-planning – under submission 2026
- **TeleChat3-MoE** training report – arXiv:2512.24157

PATENTS

Three patent families on execution-plan generation, load balancing, and multi-dimension parallelism configuration for large language model training.

EDUCATION

Sorbonne University & EURECOM 2023 – end 2026
Ph.D. in Computer Science — hybrid-parallel training optimisation; adv. Prof. Appuswamy & Prof. Li

Sorbonne University 2019 – 2022
M.Sc. in Computer Science

Paris-Saclay University 2016 – 2019
B.Sc. in Computer Science

AWARDS

- Golden Prize, Huawei Challenge Award for Automatic Load Balancing
- Best Poster Award, Euro-Par conference
- Letter of Appreciation, China Telecom

SKILLS

Parallelism: data, tensor, pipeline, virtual-pipeline, sequence, expert, optimiser

Optimisation: ILP / MILP, constraint programming, symbolic cost & peak-memory modeling

Programming: Python, C++, Java

ML frameworks: MindSpore, PyTorch

Platforms: large-scale GPU/NPU clusters (to 16k), cloud

Tooling: Ascend profiling, tracing, regression benchmarking; Linux, Git

Languages: Chinese (native), English (fluent), French (fluent), Cantonese